

October Crossword

Kevin Chen, Wenhao Lu, Aaron Qin

1	2
3	

ONE DOWN

Twelve distinct candies are distributed between children Aaron, Kevin, and Wenhao. Let N be the number of distributions in which children Kevin and Wenhao receive the same parity of candies (i.e., both get an even number or an odd number). Find the last two digits of N .

THREE ACROSS

Ten students are standing in a circular formation for a costume contest. Each student wears one of three unique costumes: Vampire, Ghost, or Zombie. No two adjacent students may wear the same costume. Let N be the number of valid circular arrangements that exist. Find the product of the (nonzero) digits of N .

ONE ACROSS

In a haunted pumpkin patch, there are 2025 pumpkins labeled $1, 2, 3, \dots, 2025$. A mischievous ghost decides to form piles of pumpkins. Each pile is a subset of the 2025 pumpkins. The total spookiness of a pile is the sum of the labels of all pumpkins in it. The ghost is only pleased with piles whose total spookiness is a multiple of 5. The number of piles that please the ghost can be written in the form $\frac{1}{m}(2^x + 2^y)$. Find m plus the sum of the digits of x and y .

TWO DOWN

A group of 64 pumpkins are arranged in a circle, each connected to its two nearest neighbors by glowing threads. At midnight, each pumpkin either glows or stays dark, independently, with equal probability $\frac{1}{2}$. After each midnight, any pumpkin that explodes is removed, and the remaining pumpkins close up to form a smaller circle, so that every surviving pumpkin is again connected to its two nearest remaining neighbors. Whenever three consecutive pumpkins all glow at the same time, the central one explodes (it disappears before the next day). Let E_n be the expected number of pumpkins that remain after n midnights. If the decimal representation of E_n terminates and has exactly 150 digits after the decimal point, determine n .